problems/MHD/fluxEmergenceSon2025

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. Read off `user_inputs.jl`, the case files and the kernels they reach on `sm/MHD`. This case pairs the conserved-form DynSGS on the *departure from a magnetostatic reference state* with a *case-local, non-conservative* density/pressure floor applied as the Runge–Kutta stage limiter. Its sibling `fluxEmergenceSon2025DSGS` drops the floor.

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	2D ideal GLM-MHD with gravity $g = 1/\gamma$, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$	Son, Jang & Magara (2025)
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2 - \frac{1}{2}\psi^2)$, $\gamma = 1.05$	`user_flux.jl`
Domain, mesh	$[0,80] \times [0,35]\, H_0$, periodic in x, walls `bottom`/`top`; `FE_80x35.msh`, 80×35 quads of $1\, H_0$	no refinement
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 4$	
Time integration	`CarpenterKennedy2N54` *with* `stage_limiter! = fe_positivity_limiter!`; fixed $\Delta t = 7.5 \times 10^{-3}\tau_0$, $t_{end} = 54\,\tau_0$	`user_inputs.jl:94, 101`
Variables	`TOTAL()`, `"energy"`	
Reference state $q_e(z)$	two-temperature magnetostatic atmosphere ($T_{ch} = 1$, $T_{cor} = 25$, $z_{cor} = 18$, $w_{tr} = 0.6$) with a flux sheet $B_x(z)$ on $[4,8]\, H_0$, $\beta_* = 1$; $\mathbf{v}_e = 0$	`initialize.jl`; `qe[:,end]` $= p_e$
Initial data	q_e plus $u = f(z)\, 0.05 \sin(2\pi(x-40)/20)$ for $\lvert x-40\rvert < 10$	paper eq. 5
Sources	`:lsource => true`: gravity $(-\rho g, -\rho v g)$, well-balanced correction of the discrete reference balance (`fe_well_balanced = true`), ψ damping, absorbing layer $z > 30$: $-\sigma(z)(q - q_e)$, $\sigma = 2\sin^2\left(\frac{\pi}{2}\frac{z-30}{z_{top}-30}\right)$	`user_source.jl`
GLM cleaning	$c_h = \max(\lvert\mathbf{v}\rvert + c_f)$ of the initial data, constant; $S_\psi = -(\alpha_p c_h/\Delta h)\psi$, $\alpha_p = 0.2$, Δh the smallest nodal spacing	paper eq. 12
Boundary conditions	walls: Dirichlet normal momentum $= 0$; `bottom` also normal $B = 0$	`user_bc.jl`
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false	

2 DynSGS: deck keys and effective values

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`, `:lvisc => true`	`SGS.jl:1598`
`:`μ	all 1	
`:dsgs_sensor`	`"legacy"`	
`:dsgs_CR`, `:dsgs_Cmax`	1.0, 0.5	
`:dsgs_Cmin`	0.03	background floor $0.03\,\Delta w$
`:dsgs_gamma`	$=\gamma_{mhd}=1.05$	
`:dsgs_Prt`	1.0	enters $\gamma(\gamma-1)/\mathrm{Pr}=0.0525$
`:dsgs_nazarov_energy`	true	split energy flux (below)
`:dsgs_norms`	`"element"`	
`:dsgs_local_rel`	1.0	
`:dsgs_conserved`	true	
`:dsgs_nodal_rho`	true in the deck, inactive	`rhs.jl:1526` forces it off under `:dsgs_conserved`
`:dsgs_ref_weight`	false (default)	
`:dsgs_fast_floors`, `:dsgs_thermal_residual`	false (defaults)	
`:dsgs_rel`	1.0 (default)	not used by element norms
`:dsgs_hold_steps`	2 (default)	
`:dsgs_cutoff`, `:dsgs_freeze_stage`, `:dsgs_reference`	0, false, false (defaults)	
`:ldsgs_nodal`	false (element form)	

3 DynSGS as it runs for this case

$\Delta_K^{elem}=1\,H_0$, $\Delta_K=\Delta_K^{elem}/(N+1)=0.2\,H_0$, $h=\Delta t$.

1. **History and hold.** Rotation once per step; two held steps: $R\equiv 0$, so $\nu=\nu_{\mathrm{fl}}=0.03\,\Delta_K w_K$ during steps 1–2.
2. **Legacy residual**, every stage: $R_I^s=\left|\frac{3q_I^s-4q_I^{n,s}+q_I^{n-1,s}}{2h}-(M^{-1}\mathcal{R}_{inv})_I^s\right|$, with the inviscid RHS (fluxes and *all* sources: gravity, well-balanced term, ψ damping, absorbing layer) assembled over ranks and the periodic seam. The Dirichlet-node zeroing exists only on the residual-sensor path, so the wall nodes are *not* zeroed here.
3. **Element normalization** (`SGS.jl:1716-1761`): over the 25 nodes of K, $\langle q\rangle_K$ (plain average), $S_K^s=\max|q^s-\langle q^s\rangle_K|$, $\bar\rho_K$, $\bar p_K$ (without ψ), $\bar c_K=\sqrt{\gamma\bar p_K/\bar\rho_K}$ (sound speed),

$$n_K^s=\max\left(S_K^s,\ r_{loc}\,\sigma_K^s\right)+10^{-16},\quad \sigma_K^\rho=\bar\rho_K,\ \sigma_K^{\mathbf{m}}=\bar\rho_K\bar c_K,\ \sigma_K^E=\bar\rho_K\bar c_K^2,\ \sigma_K^{\mathbf{B}}=\sqrt{\bar\rho_K}\,\bar c_K,$$

   $r_{loc}=1$. Each element is measured against its own scales, so the corona ($\rho\sim 10^{-8}\rho_0$) is not hidden by the photosphere.
4. **Ratio, cap, coefficient.** $\mathcal{R}_K=\max_{I\in K}\max_{s\le 8}R_I^s/n_K^s$; $w_K=\max_I(|\mathbf{v}_I|+\sqrt{\max(\gamma p_I/\rho_I+|\mathbf{B}_I|^2/\rho_I,0)})$;

$$\nu_c=\max\left(0,\min(0.5\,\Delta_K w_K,\ \Delta_K^2\mathcal{R}_K)\right),\qquad \nu_{\mathrm{fl}}=0.03\,\Delta_K w_K,\qquad \nu_K=\max(\nu_c,\nu_{\mathrm{fl}}),$$

   one value per element, recomputed at every stage.
5. **Slot coefficients** (conserved branch with `lnazarov_energy`, `SGS.jl:1881-1885`): ν_K on slots $1,2,3,5,\ldots,9$; slot 4 gets $\kappa_K^E=\max\left(\gamma(\gamma-1)/\mathrm{Pr}\cdot\nu_c,\ \nu_{\mathrm{fl}}\right)=\max(0.0525\,\nu_c,\nu_{\mathrm{fl}})$.

4 The viscous operator

`rhs.jl:1533` sets `dsgs_split_energy[] = true`; `user_primitives.jl` then returns the departure from q_e with the energy split into its non-thermal and thermal parts, $\delta q=q-q_e$, $E_{th}=p/(\gamma-1)$, $E_{nth}=E-E_{th}$, the latter in the spare slot $n_{eqs}+2=11$ of `uprimitive`. `_expansion_visc!` (`rhs.jl:3050-3066`) assembles

Equation	Viscous flux $\mathbf{F}_V$	Remark
ρ	$\nu\nabla\delta\rho$	mass diffusion on
$\rho u, \rho v, \rho w$	$\nu\nabla\delta(\rho u), \ldots$	scalar Laplacians, no stress
E	$\nu\nabla\delta E_{nth} + \kappa^E\nabla\delta E_{th}$	$\nu =$ the ρu-slot coefficient
B_k	$\nu\nabla\delta B_k$	$\eta = \nu$
ψ	$\nu\nabla\psi$	$\psi_e = 0$

A Laplacian of the conserved state (here: of its departure from q_e), in divergence form, with no $\tau\cdot\mathbf{v}$ or resistive-work term. At rest ($q = q_e$) the operator vanishes.

5 Positivity

Case-local floor as the RK stage limiter (`fe_positivity_limiter!`, `user_inputs.jl:29-77`)

This is *not* the `:lpositivity` machinery of `src/kernel/positivity` (`:lpositivity` stays `false`); it is a function of the deck handed to the solver as `CarpenterKennedy2N54(stage_limiter! = fe_positivity_limiter!)`. Jexpresso's `solve` passes a non-empty `saveat`, so OrdinaryDiffEq sets `calck = true`, which forces `williamson_condition = false`: the floor runs before the RHS of stages 2–5 and after stage 5. Per node I, with $f = 0.2$ (`fe_floor_frac`) and the reference values at that node:

1. if $\rho_I < f\rho_{e,I}$: $\rho_I \leftarrow f\rho_{e,I}$ (momentum unchanged, i.e. the velocity is reduced);
2. with $k = |\mathbf{m}_I|^2/(2\rho_I)$ (new ρ) and $m = \frac{1}{2}|\mathbf{B}_I|^2 + \frac{1}{2}\psi_I^2$: if $(\gamma-1)(E_I - k - m) < f\,p_{e,I}$, then $E_I \leftarrow f\,p_{e,I}/(\gamma - 1) + k + m$ ($\mathbf{m}$, $\mathbf{B}$, ψ unchanged).

It **adds mass and energy** wherever it fires (not conservative). Hits are counted per rank and reported at the first hit and each further decade, with a sample (x, z); the count is rank-local (printed with the `MPI.COMM_WORLD` rank).

Other ingredients

- Flux-only floor: $p \leftarrow \max(p, 10^{-9})$ inside `user_flux!` (`p_floor_mhd`); the state is untouched.
- The DynSGS mass diffusion and the $C_{\min}$ background act in the same direction.

6 Observations: code versus comments

1. "Slot 4 carries the TOTAL ENERGY: `"energy"` keeps the kernel's $\tau\cdot u$ … active": under `:dsgs_conserved => true` the $\tau\cdot\mathbf{v}$ term is switched off (`rhs.jl:2965`).
2. "$\Delta t = 2.5\times10^{-3}\tau_0$ is CFL ≈ 0.07": the deck sets 7.5×10^{-3}.
3. The floor reaches stages 2–5 only because `calck = true` in Jexpresso's `solve`; the deck does not set `williamson_condition = false` as the conservative limiter does (`positivity_driver.jl:266-268`). A run without callbacks or `saveat` would apply it once per step.

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	**Where (file:line)**
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report	`src/io/mod_inputs.jl:456-477`
`:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H)+1$ rotations)	`rhs.jl:182`

Mechanism	**Where (file:line)**
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\,\mathbf{B} \cdot \nabla\mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs}+2$ in 2D, $n_{eqs}+1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)